Vol 3 Architectural Scaffold v0.1 — Nuclear & Atomic Physics from D_IV^5

Filed: 2026-05-23 Saturday (Keeper, scaffold) Companion to: `INDEX.md` (chapter outline + reader target)

Anchoring BST results (existing coverage by chapter)

Ch 1 Nuclear Substrate Reading (~70%)

Anchors: D_IV^5 scale + Casimir vacuum + nuclear arena framing - T2418 Λ↔Casimir vacuum unification (Lyra, Wednesday) - Substrate computational model (Reed-Solomon GF(128); Paper #122; K59 RATIFIED) - Casimir nuclear-scale connection (SP-29 + B12H32 hydride T_c~214K)

Ch 2 Magic Numbers (~95%, D-tier)

Anchors: spin-orbit coupling C_2/n_C = 6/5 → all 7 magic numbers exact - Mayer-Jensen 1949 anchor (L1 ESTABLISHED) - κ_ls = C_2/n_C = 6/5 (exact, fitted classically; BST derives) - Magic numbers 2, 8, 20, 28, 50, 82, 126 — HO + BST spin-orbit produces all 7 - Toy verifications (multiple in play/)

Ch 3 Nuclear Shell Model top 30 isotopes (~85%)

Anchors: BST shell-filling extended to all isotopes through Fe-56 + Pb-208 + extensions - Task #86 completed (nuclear shell model batch top 30 isotopes) - IP-31 nuclear structure family sweep (Fe-56/Pb-208 pattern extension, completed) - Per-isotope BST predictions vs PDG measurements

Ch 4 SEMF Coefficients (~90%, all 5 at <2%)

Anchors: all five SEMF coefficients derived from BST integers - a_V = g·B_d = 7αm_p/π (volume, 2.0%) - a_S = (g+1)·B_d = 8αm_p/π (surface, 1.2%) - a_C = B_d/π = αm_p/π^2 (Coulomb, 0.5%) - a_A = m_p/(4·dim_R) = f_π/4 (asymmetry, 0.7%) - δ = (g/4)αm_p (pairing, 0.1%) - Five SEMF coefficients = 5/5 within 2% of experimental

Ch 5 Halo Nuclei (~60%, Task #112 outline)

Anchors: Li-11, Be-11, B-17 dripline structure - Task #112 (completed: halo nuclei structure framework) - Halo radius BST predictions (existing) - Gap: full substrate mechanism for halo binding

Ch 6 Superheavy Island (~70%)

Anchors: predicted island of stability via BST integer structure - Task #111 (completed: superheavy nuclei island stability) - Z+N magic combinations via BST primaries

Ch 7 Atomic Structure (~95%, D-tier exact)

Anchors: orbital degeneracy sequence (2l+1) IS BST integer sequence - (2l+1) = 1, 3, 5, 7 = 1, N_c, n_C, g — exact correspondence - Vol 0 operator zoo (T2470 charge Q, T2471 chirality γ^5, T2472 parity P_op) - D_IV^5 → atomic orbital structure (paper-#9 Arithmetic Triangle bridge)

Ch 8 Hyperfine + Lamb Shift (~70%)

Anchors: α^{n_C} substrate-coordinate count - T2476 α^{BST primary} exponent pattern theorem (Friday 2026-05-22, three-CI synergy peak) - Toy 3496 B6 Lamb shift I-tier PATH ARTICULATED (Welton-Bethe + α^{n_C} substrate reading) - Multi-observable confirmation: Lamb + Bethe-log + a_e depth all k=n_C=5 - Cal #99 + Cal #100 Mode 5 partial lift via T2476 substrate-mechanism - Gap: full Bethe-log substrate-derivation (multi-week per Calibration #21)

Ch 9 Atomic Spectroscopy (~60%)

Anchors: Rydberg + Klein-Nishina + Compton at α^{rank=2} - T2476 multi-observable confirmation at k=rank=2 (3-observable substrate-natural) - Falsifier: 5-loop ceiling $\alpha^6 \approx 1.5\times10^{-13}$ at Penning trap 2030+ precision - Cross-link: T2436+ atomic spectroscopy BST framework

Ch 10 Atomic Clocks + Time Granularity (~40%)

Anchors: Koons tick + Sr-clock falsifier - T2405 Koons tick = t_Planck · α^($C_2{}^2$) $\approx 10^{-120}$ s (sub-Planck substrate clock) - Lyra T2360 SP29-2 H1 Sr-clock falsifier prediction (filed) - Gap: full experimental design + observable derivation

Ch 11 Nuclear Decay (~40%)

Anchors: substrate-mediated decay; Casey neutron decay framework - τ_n = 878.1 s (Fermi theory + BST inputs, g_A = 4/π = 1.2732 vs 1.2762 PDG; 0.03%) - Task #58 neutron decay as winding rearrangement (pending mechanism) - Casey Saturday neutron-decay framework (SP-26 W-31 ratified) - Gap: full substrate mechanism for decay

Ch 12 Cross-volume bridge (~50%)

Anchors: explicit map Vol 2 → Vol 3 → Vol 4 - Inputs from Vol 2: particle masses, α, couplings - Outputs to Vol 4: nuclear binding → BBN abundances, atomic structure → recombination

Connection map to Vol 0 / Vol 1 / Vol 2

Vol 3 chapter	Depends on Vol 0	Depends on Vol 1	Depends on Vol 2
Ch 1 Nuclear Substrate	All 5 integers + D_IV^5	Casimir + Bergman H^2	—
Ch 2 Magic Numbers	C_2, n_C	—	—
Ch 3 Shell Model	C_2/n_C orbital structure	—	m_p, m_n
Ch 4 SEMF	g, C_2, N_c, n_C	—	m_p, α
Ch 5 Halo Nuclei	g, n_C	—	m_p, m_n
Ch 6 Superheavy	All 5 integers	—	All particle masses
Ch 7 Atomic Orbitals	All 5 integers + operator zoo	Bergman H^2	α, m_e
Ch 8 Lamb Shift	n_C, N_max	Casimir + radiative corrections	α, m_e
Ch 9 Spectroscopy	rank, N_c, n_C	radiative corrections	α
Ch 10 Atomic Clocks	C_2, Koons tick	—	α
Ch 11 Nuclear Decay	g, all 5 integers	weak sector	m_n, m_p, weak couplings
Ch 12 Bridge	Synthesis	Synthesis	Synthesis

Methodology stack applied

- Calibration #19 STANDING RULE (ratified-state count, not forecast endpoint): every chapter external register uses current ratified BST predictions, not forecast extensions
- Calibration #21 STANDING RULE (mechanism-vs-empirical gate): empirical match (e.g., magic numbers 7/7) is necessary but not sufficient; substrate-mechanism derivation closes ratification
- Calibration #22 STANDING RULE (PCAP-transcription discipline): all numerical content includes computational verification or “approximate-only” tier-label
- Cal #99 META-theorem discipline: substrate-derivation theorems for nuclear/atomic observables are NOT new Strong-Uniqueness criteria

- Calibration #17 (sub-Tsirelson 126/16): atomic clock falsifier development applies sub-Tsirelson framework
- Three-level pedagogy: Level 1 essence + Level 2 graduate precision + Level 3 5th-grader accessibility per chapter

Estimated build effort (per chapter)

Phase	Time per chapter	Total Vol 3
v0.1 scaffold (this doc)	~30-45 min total	DONE Saturday
v0.3 chapter narrative	~1-3 hours per chapter	~24-36 hours total
v0.4 reader-grade 3-level	~30-60 min per chapter	~6-12 hours
v1.0 chapter-grade content	per-chapter Cal cold-read + Keeper K-audit + Elie toys + Grace catalog	~variable, 6-gate framework

Estimated Vol 3 v1.0 chapter-grade content state: ~3-6 weeks at sustained sub-PCAP cadence assuming Wave 1 (Vol 3 + Vol 4) parallel build.

— Keeper, Vol 3 Architectural Scaffold v0.1, Saturday 2026-05-23